Phantom Non-Identifiability: When Simplifying a Coupled Simulator Fabricates an Observability Boundary

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Abstract

Reduced simulators are the substrate of physics-informed surrogates and simulation-based inference, and identifiability and optimal-sensor analyses are increasingly run on the same reduced model. We show that a reduction severing the only path from a parameter to an observed field does not merely lose accuracy: it manufactures a structural non-identifiability the real system does not have, together with a confident prescription to buy an instrument the real system may not require for identifiability. We demonstrate the mechanism on coupled heat and moisture (Luikov) transport, where dropping the thermogradient (Posnov) coupling makes a Fisher analysis declare the moisture sensors blind to the temperature-side parameters and an E-optimal step prescribe a temperature channel. Restoring the term collapses the null: sensitivities grow as $O(\mathrm{Pn})$ and the Cramér-Rao bound as $O(\mathrm{Pn}^{-1})$, which we derive and use to *predict* the crossover from one probe solve. Critically, the failure is silent – over a wide window the calibrated reduced model passes a χ^2 goodness-of-fit test while still reporting rank 2 and "buy a temperature sensor" – and it is not confined to inference: the same severed path zeros the input-to-observable Jacobian, so a controller built on the reduced simulator pays up to $12.5\times$ the matched controller's closed-loop cost while the objective component its own model treats as controllable improves. A frozen parametric PINN inherits the phantom, and a two-head control rules out the obvious confound. We give the check to run on other reductions.

1 Introduction

The mechanism. A reduced simulator is safe for a task only if it preserves the directed dependencies that task differentiates through. Inference and sensor design differentiate the observation operator H with respect to parameters θ; control differentiates it with respect to inputs u. *Severing the only path from a parameter to an observable is **sufficient** to create an exact null.* Neither converse holds: a surviving path may be weak, wrong-signed or cancelling, and a severed one need not be fatal if the task's Jacobian is carried by another route – so the rule we draw is to check the relevant tangent map, not to audit path structure.[1] Sever the only path and $\partial H/\partial\theta_j \equiv 0$ identically – not small, zero – so the reduced model reports the parameter as structurally unidentifiable and an optimal-design step confidently prescribes new instrumentation to repair a deficiency the system does not have.

Reductions of the same shape recur across multiphysics modelling, and we offer them as candidates where the check below is worth running rather than as established instances: *one-way coupling* in fluid–structure interaction and particle-laden flow – a phrase that sounds like a documented

[1] "Observability" here means local parameter identifiability (full column rank of the scaled sensitivity, a finite Cramér-Rao bound) plus input-to-observable sensitivity, not classical state observability or controllability.

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approximation but is a structural claim about what can be inferred from what – and likewise neglected Soret or Dufour cross-diffusion, a zero Biot coefficient in poroelasticity, or frozen chemistry in a reacting flow. Each severs a specific dependency.

This paper, and how we found it. The system is coupled heat and moisture transport in a drying sheet (the Luikov equations [12]), where a fast mesh-free surrogate is wanted as a soft sensor for closed-loop control; PINNs [18] fit, and such systems are increasingly modelled this way [13, 10]. Building one forces the identifiability question – which parameters are recoverable from moisture-side sensors alone, and which extra instrument is needed? – now routinely posed on the same simplified simulator the surrogate is trained on [4]. We built exactly that, and wrote it up: the parametric PINN on the reduced ($\mathrm{Pn}{=}0$) model, the Fisher analysis on it, the rank-2 null on $\{\mathrm{Ko}, \mathrm{Bi}_t\}$, and the E-optimal "add a temperature channel" prescription. A pre-specified check of the dropped term, with thresholds fixed before the sweep was run, was meant to confirm that result; it dissolved it. Sec. 3 is the original analysis, unchanged;[2] the rest is the correction. We show the frozen PINN inheriting the phantom; by the same argument any estimator trained only on the reduced simulator should inherit it, amortized SBI and neural operators included, though we do not demonstrate those – so a leaderboard-accurate surrogate can encode a confident but wrong observability verdict, the misspecification regime this venue targets [4].

Related work. Identifiability, Cramér-Rao bounds and optimal experimental design are classical [7, 17] and central to SBI under misspecification [4]. For Luikov drying the inverse-problem literature poses estimation and design on *temperature* data – by a D-optimum criterion over heat flux, timings and sensor placement [5], on the temperature field as sensitivity state variable [11], or by a heat-flux profile making all parameters simultaneously estimable [2] – and none asks what a *moisture-only* instrument can identify. Drying surrogates use PINNs on the reduced model [18, 13, 10]; discrepancy-corrected and universal PINNs learn missing operators [19, 16] and derivative-informed surrogates preserve high-fidelity Jacobians [14], but these target state correction or assume derivative labels. Closest are the literatures on model error itself: discrepancy-aware Bayesian calibration, where ignoring discrepancy biases physical parameters and overstates confidence [8, 3], and optimal design under model error, irreducible [9] or reducible [1]. All place the discrepancy in the *outputs*, leaving the tangent map intact; here the reduction zeros $\partial H/\partial\theta_j$ exactly, which no output-space term can express, so the design verdict flips while the fit stays good.

Contributions. **(i)** The moisture-only null exists only at $\mathrm{Pn}{=}0$, and we *derive* rather than measure its dissolution. **(ii)** The failure is *silent*: a calibrated reduced model passes a χ^2 goodness-of-fit test while returning the phantom null. **(iii)** The sweep becomes a *predictive* diagnostic from one probe solve, whose distribution over the box we report by a Sobol design. **(iv)** A two-head control separates inherited physics from a dead input, and no tested loss or architecture restores it. **(v)** The same severed path removes the actuator's authority in closed loop. Throughout we validate the chain end to end – physical mass-flux operator, per-channel likelihood, Cramér-Rao against Monte Carlo, a withheld instrument, and the joint five-parameter problem, which we report as *not* well posed. Code, the JSON artefact behind every number, and this paper's source: `https://github.com/codebymov/phantom-non-identifiability`.

2 Setup

Simulator. On $\xi \in [0,1]$ (through-thickness, $\xi{=}0$ the hot-air surface) and dimensionless time τ, the Luikov system with the thermogradient coupling is [11, 12]

$$\partial_\tau T = \partial_{\xi\xi} T + \mathrm{Ko}\,\partial_\tau M, \qquad \partial_\tau M = \mathrm{Lu}\,\partial_{\xi\xi} M + \underbrace{\mathrm{Lu}\,\mathrm{Pn}\,\partial_{\xi\xi} T}_{\text{thermogradient (Posnov)}}, \tag{1}$$

with Robin conditions at $\xi{=}0$ ($\partial_\xi T{=}\mathrm{Bi}_t(T{-}1)$, $\partial_\xi M{=}\mathrm{Bi}_m M$), Neumann at $\xi{=}1$, and $T(\xi,0){=}0$, $M(\xi,0){=}1$. The standard reduction sets $\mathrm{Pn}{=}0$, making the moisture balance autonomous; this is the form used across drying surrogates [13, 10]. Ground truth is an implicit finite-difference (FD) solver extended to carry Pn (default 0, bit-identical to the reduced solver). The canonical Luikov benchmark takes $\mathrm{Pn}{=}0.6$ [11] and $[0.2, 2]$ is the modelling range for hygroscopic capillary-porous

[2] Except that its temperature channel is relabelled to the operator actually used, the mean $\langle T\rangle$; no result changed.

media [12]; assuming paper sits in that class, we call it the *assumed physical range* – an assumption, not a measurement.

Observation operator. A drying cell instruments two moisture quantities: a gravimetric scale giving $\bar{M}(\tau) = \langle M(\cdot,\tau)\rangle_\xi$, and a mass-flow meter giving the surface mass flux. In conservation form $\partial_\tau M = -\partial_\xi q_m$ with $q_m = -\mathrm{Lu}(\partial_\xi M + \mathrm{Pn}\,\partial_\xi T)$, so the flux is the correct mass-flow operator and, with the Neumann back boundary, $q_m(0,\tau) = \mathrm{d}\bar{M}/\mathrm{d}\tau$ exactly. The Robin conditions give both boundary derivatives in closed form, so no numerical differentiation enters:

$$q_m(0,\tau) = -\mathrm{Lu}\big(\mathrm{Bi}_m M(0,\tau) + \mathrm{Pn}\,\mathrm{Bi}_t(T(0,\tau)-1)\big). \tag{2}$$

They remain separate instruments with independent noise; stacked over $K{=}40$ sample times they form H_M, with $\langle T\rangle{=}\langle T(\cdot,\tau)\rangle_\xi$ the optional addition – used for the gain row of Table 1 and the E-optimal step of Sec. 3, while App. B withholds the surface trace $T(0,\tau)$. On the grid used the identity holds discretely to 0.4% for $\mathrm{Pn} \le 0.2$, degrading to 2.6% at $\mathrm{Pn}{=}2$ as the explicit thermogradient term grows; it doubles as a conservation check.

Noise model and identifiability. Each channel carries independent Gaussian noise with σ_c equal to 1% of its own range. We form the scaled sensitivity $S_{ij} = \partial H_i/\partial \log\theta_j$ for $\theta = (\mathrm{Lu}, \mathrm{Ko}, \mathrm{Bi}_t, \mathrm{Bi}_m)$ by central differences through the solver ($d{=}10^{-3}$ in $\log\theta$), the Fisher information $F = S^\top\Sigma^{-1}S$, and the Cramér-Rao (CR) relative standard deviation [7]. Because Ko and Bi_t enter only the temperature equation, *at* $\mathrm{Pn}{=}0$ *the moisture balance is autonomous and Eq. 2 collapses to* $-\mathrm{Lu}\,\mathrm{Bi}_m M(0,\tau)$*, so* $\partial H_M/\partial\mathrm{Ko} = \partial H_M/\partial\mathrm{Bi}_t \equiv 0$*,* $\mathrm{rank}(S) = 2$*, and one mean-temperature channel restores full column rank.* That one-line fact is the whole of the reduced-model result; the contribution is what happens when the dropped term returns. Rank is read from the scaled columns, not an eigenvalue ratio – column j counts when $\|S_{:j}/\sigma\| > 10^{-9}\max_k\|S_{:k}/\sigma\|$ – so weak-but-nonzero coupling is reported by the CR bound, never as a spurious null.

3 What the reduced simulator prescribes ($\mathrm{Pn}{=}0$)

At the nominal $\theta{=}(0.5, 2, 1, 1)$ the moisture-only $\{\mathrm{Ko}, \mathrm{Bi}_t\}$ sensitivity columns are exactly zero and $\mathrm{rank}(S) = 2$. On the reduced model this reads as a firm design law: an E-optimal ($\lambda_{\min}(F)$) selection [17] lifts $\lambda_{\min}$ from 0 to $3.2{\times}10^2$ by adding a mean-temperature channel, matching the temperature-first prescription of the Luikov literature [5, 2]. The reduced simulator hands the experimenter a confident verdict: moisture instrumentation is structurally insufficient. Three estimators agree, the misfit being exactly flat in those coordinates: solver-in-the-loop least squares and an amortized estimator fitting the $(\mathrm{Ko}, \mathrm{Bi}_t)$ inputs of a frozen parametric PINN never leave their *initialisation*, and ensemble-Kalman inversion [6] – the only one with a prior in the Bayesian sense – does not contract. The surrogate's architecture, training budget and forward accuracy are in App. A.

Control: inherited physics, or a dead input? A vanishing derivative with respect to a network input is equally the signature of a coordinate never learned, so the inheritance claim and a training failure are indistinguishable as usually stated. They separate without retraining, because the same network has two heads and Ko reaches them by different routes: directly in the temperature equation, and into the moisture equation *only* through the dropped term. Autograd through the frozen surrogate gives $\|\partial\langle T\rangle/\partial\log\mathrm{Ko}\| = 2.498$ against the solver's 2.485 (1.2%) while $\|\partial\langle M\rangle/\partial\log\mathrm{Ko}\| = 0.022$ against its exact zero (Bi_t behaves the same). A dead input would zero both heads. One caveat: the moisture-head residue is not exactly zero, and at roughly 7% of the true coupled sensitivity at the best-converged arm (App. C) it is small but not negligible – a surrogate's tangent map can report a coupling as small, and cannot certify it as zero. Nor is it an objective one can fix: gradient-enhanced penalties, gradient-norm balancing, causal weighting and a shared trunk feeding T and $\partial_\xi T$ straight to the moisture head all drive the severed sensitivity *toward* zero rather than toward its true coupled value, over three seeds; the architecture built to let the pathway reappear is indistinguishable from objectives with no such design (App. C).

4 The boundary dissolves, and the rate is derivable

The reduction is not free: the full moisture balance carries the thermogradient term $\mathrm{Lu}\,\mathrm{Pn}\,\partial_{\xi\xi}T$ [11]. Expanding about the decoupled solution, $M = M_0 + \mathrm{Pn}\,M_1 + O(\mathrm{Pn}^2)$ with $\partial_\tau M_1 = \mathrm{Lu}\,\partial_{\xi\xi}M_1 +$

$\mathrm{Lu}\,\partial_{\xi\xi}T_0$, both channels inherit $\mathrm{Ko}, \mathrm{Bi}_t$ only through T_0 at first order (the flux also through the explicit $\mathrm{Pn}\,\mathrm{Bi}_t$ term of Eq. 2, likewise $O(\mathrm{Pn})$). Hence

$$\partial H_M/\partial \log\theta_j = \mathrm{Pn}\,a_j(\theta) + O(\mathrm{Pn}^2), \qquad F_{jj} = \mathrm{Pn}^2 a_j^2/\sigma^2, \qquad \mathrm{CR}(\theta_j) = \gamma_j(\theta)/\mathrm{Pn} + O(1). \tag{3}$$

Over $\mathrm{Pn} \in [10^{-3}, 10^{-1}]$ the log-log slopes of the two CR bounds are -0.959 and -0.990 against -1 from Eq. 3, and the CR falls through the 100% usability line at a derived rather than measured rate (Fig. 1a, Table 1). Two statements must be kept apart. *Analytically*, Eq. 3 removes the exact null wherever $a_j(\theta) \neq 0$ – the generic case, since T_0 carries Ko and Bi_t and the restored $\mathrm{Lu}\,\mathrm{Pn}\,\partial_{\xi\xi}T_0$ is not identically zero – but does not exclude isolated θ where some a_j vanishes. *Numerically*, $\mathrm{rank}(S)=4$ at every tested $\mathrm{Pn}>0$, over the box and the $K=40$ schedule. Identifiability is graded, not binary, in three regimes: exactly zero columns at $\mathrm{Pn}=0$; nonzero but nearly parallel to the $\{\mathrm{Lu}, \mathrm{Bi}_m\}$ columns for $0 < \mathrm{Pn} \lesssim 0.7$ ($|\mathrm{corr}(\mathrm{Ko}, \mathrm{Bi}_t)|=0.997$ at $\mathrm{Pn}=0.2$), where the marginal CR stays large through collinearity rather than a null; and genuinely separated above $\mathrm{Pn} \approx 0.8$, where those correlations collapse and the marginal CR falls steeply (Fig. 1a).

A predictive diagnostic, not a sweep. Eq. 3 makes the crossover $\mathrm{Pn}^\star$ (where $\mathrm{CR} = 100\%$) equal to $\gamma_j(\theta)$, which one probe solve estimates as $\mathrm{CR}(\theta_j; \mathrm{Pn}_{\text{probe}}) \cdot \mathrm{Pn}_{\text{probe}}$. Against bisection over the 16 box corners this is accurate to 4.5% median and 15.5% max where the crossover is itself weakly coupled ($\mathrm{Pn}^\star < 0.05$), degrading to 18.7% median where $\mathrm{Pn}^\star = O(1)$ – a failure the probe itself flags. Comparing $\mathrm{Pn}^\star$ with the plausible range of the dropped coefficient is the operational test: below that range the null is a phantom, above it the reduced model's verdict stands. The test settles the *structural* claim – identifiability in principle – not whether the instrument is worth buying. It presumes the dropped term is known and restorable with a plausible range, and $\mathrm{Pn}^\star$ inherits the 100% threshold, noise model, schedule and nuisance choice: a check for a *suspected* coupling, not a detector of unknown structural error.

Crossover distribution. Over 256 Sobol points the crossover has median $\mathrm{Pn}^\star=0.037$ (interquartile $[0.019, 0.089]$), and 88% of the box lies below 0.2, the bottom of the assumed physical range: for most operating conditions $\{\mathrm{Ko}, \mathrm{Bi}_t\}$ cross the 100% CR line at realistic coupling *with* Pn *known* – identifiability in principle, not precision, since below $\mathrm{Pn} \approx 1$ they stay weakly determined (App. B). The remaining 12% (low Lu and Ko) never crosses it.

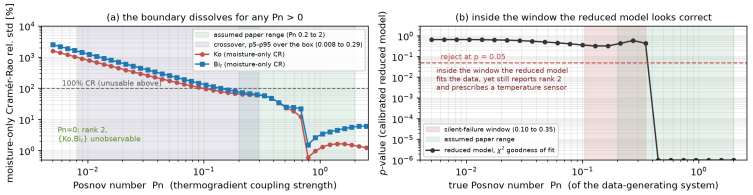


Figure 1: **(a)** Moisture-only CR relative std for $\{\mathrm{Ko}, \mathrm{Bi}_t\}$ vs. the Posnov coupling: unidentifiable at $\mathrm{Pn}=0$, full rank at every tested $\mathrm{Pn}>0$. **(b)** The failure is silent: the reduced model, calibrated over $(\mathrm{Lu}, \mathrm{Bi}_m)$ against the same noisy moisture data, passes a χ^2 test up to $\mathrm{Pn}=0.35$ while returning the phantom null throughout. Between the crossover (0.0996 at the nominal θ) and the top of that band, nothing signals that anything is wrong.

Table 1: Moisture-only identifiability vs. Posnov number (nominal θ, $K=40$, per-channel noise 1% of range). Gain is the moisture-only to moisture-plus-$\langle T \rangle$ CR ratio; $\mathrm{rank}(S)=2$ at $\mathrm{Pn}=0$, 4 at every tested $\mathrm{Pn}>0$.

Pn	0.0	0.05	0.1	0.2	0.6	1.0
CR Ko	∞	176%	99%	66%	19%	1.2%
CR Bi_t	∞	259%	134%	77%	26%	3.2%
gain from a T channel (Ko)	∞	34×	14×	5.1×	8.1×	1.2×

5 The failure is silent

At $\mathrm{Pn}{=}0.6$ the reduced model at the true parameters leaves a 17% moisture discrepancy, which would be caught before its verdict was trusted. But a practitioner calibrates first, and calibration launders the misspecification into the parameters it *can* see. We generate noisy moisture-only observations from the full model at each true Pn, fit $(\mathrm{Lu}, \mathrm{Bi}_m)$ under the reduced model in the operating box, and test the residual against the known per-channel noise (χ^2, 20 seeds, $2K{-}2$ d.o.f.). It passes at $\alpha{=}0.05$ up to $\mathrm{Pn}{=}0.35$ (Fig. 1b), where the median p-value is 0.44 and the calibrated discrepancy 0.2%, down from 11.9% – the error absorbed as bias in Lu, pulled from 0.5 to 0.33, while the Jacobian rotates and the sensitivity ordering inverts (App. B). Throughout, the reduced model reports $\mathrm{rank}(S) = 2$, $\mathrm{CR} = \infty$ and "buy a temperature sensor". The window runs from $\mathrm{Pn}^\star{=}0.0996$ to 0.35, overlapping the assumed range on $[0.2, 0.35]$: goodness of fit misses the problem because the misspecification is invisible in the observables and lethal only in the Jacobian.

6 OSSE: recovery, robustness, and the joint problem

The Fisher verdict is estimator-agnostic; we confirm it with an estimator. From the two noisy moisture-only channels we recover the "unobservable" pair by solver-in-the-loop least squares in log-parameter space, from a wrong initialisation, under either forward model (Table 2). Under the reduced model the estimator is frozen with zero variance across seeds and at every noise level from 0.2% to 5%: the misfit is exactly flat, so the pair is *structurally impossible* to recover. Under the full model the *same* channels recover both to 0.1% (Table 2). Under a genuine instrument misspecification (unmodelled 3% gain, 0.5% bias) Ko degrades to 12.7%, which the temperature channel only improves to 9.6%.

Validation, and the limit of the claim. The Cramér-Rao bound agrees with the estimator's Monte Carlo spread once compared like with like, and a moisture-only fit predicts a withheld surface-temperature channel the reduced model cannot (App. B). But local full rank is not global identifiability: a five-parameter recovery from random initialisations leaves $\mathrm{Bi}_t, \mathrm{Bi}_m$ at 45% and 41%, in more than one basin. The dissolution concerns the $\{\mathrm{Ko}, \mathrm{Bi}_t\}$ null, not the well-posedness of the joint problem: with Pn known that is unimodal but well determined only at the strongest coupling we tested, $\mathrm{Pn}{=}1$ (App. B).

7 Implications, mitigation, and conclusion

Inheritance and a model-uncertain correction. Estimators trained on the reduced simulator cannot recover a parameter it made invisible, whatever their forward accuracy. Fixing $\mathrm{Pn}{=}0$ is the degenerate structural prior $p(\mathrm{Pn}){=}\delta_0$; a spike-and-slab instead returns uncertainty over $J = \partial H/\partial\theta$ and the sensor ranking, and tracks the matched controller for $\mathrm{Pn} \geq 0.35$, at a cost where the coupling is weak (App. E). What we invalidate is the *structural* argument – that moisture instrumentation is categorically insufficient – not the instrument: under Pn uncertainty a temperature channel still informs Pn, breaks the collinearity and cuts the bound at every tested $\mathrm{Pn}{>}0$ (Table 1). Nor is fitting observations enough, the object that must be right being the tangent map: a discrepancy network should learn a conservative cross-field *flux* with parameter–discrepancy separation [15], checkable by our two-head control and set out as a three-step audit in App. D.

Phantom uncontrollability. The severed path corrupts intervention as well as inference. With the hot-air temperature u manipulated, the reduced plant advances moisture by an operator referencing neither u nor T: $\partial\bar{M}/\partial u \equiv 0$ to machine precision while the actuator keeps full authority over temperature, returning at first order like the parameter Jacobian. The consequence is a decision, not a bound: at $\mathrm{Pn}{=}0.35$, inside the silent band, a reduced-model controller pays $8\times$ the matched controller's cost while the component its own model treats as controllable improves (App. E).

Limitations and the lesson. All results are in silico from one solver family and one coupled system; a second system and output feedback are future work. The Posnov range is assumed. The silent band uses one noise model, schedule and box, with its χ^2 test on the calibration data and the two fitted parameters removed from the d.o.f. The lesson: validating a reduced simulator's *outputs* is insufficient when inference or control depends on tangent maps it has severed.

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   Justification: The structural claim (Sec. 2) is a one-line consequence of the moisture balance being autonomous at $\mathrm{Pn}{=}0$, with its assumptions stated alongside it and verified numerically (columns exactly zero, $\mathrm{rank}(S){=}2$). The scaling law (Eq. 3) is a first-order expansion about the decoupled solution, valid for $\mathrm{Pn} \ll 1$; its predicted exponents are checked against measurement (-0.959, -0.990 vs -1) and its predictive use is reported separately inside and outside its regime of validity. Sec. 4 separates what the expansion proves – no exact null wherever $a_j(\theta) \neq 0$, which does not exclude isolated θ at which some a_j vanishes – from the numerical finding of full rank at every *tested* $\mathrm{Pn}{>}0$ over the box and the $K{=}40$ schedule, and Sec. 2 states the scaled-column-norm criterion by which rank is decided.

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A Surrogate details

The amortized estimator of Sec. 3 fits the $(\mathrm{Ko}, \mathrm{Bi}_t)$ inputs of a frozen parametric PINN: two MLPs (width 96, depth 6, hard-enforced initial conditions) over $(\xi, \tau, \mathrm{Bi}_t, \mathrm{Bi}_m, \mathrm{Lu}, \mathrm{Ko})$, trained physics-only on 8000 collocation points for 12k Adam then 1.5k L-BFGS iterations. Forward accuracy at the nominal point is rel-L_2 0.012 on $\langle M \rangle$ and 0.111 on $\langle T \rangle$. The temperature head's field error being 11% while its Ko-sensitivity matches the solver to 1.2% is not a contradiction: a network can misplace a field's level and still capture how that field responds to a parameter, and it is the latter that the two-head control of Sec. 3 relies on to establish that the zero moisture-side sensitivity is inherited physics rather than a training artefact.

B Validation detail

Table 2: OSSE recovery of the "unobservable" pair from the two moisture channels (truth $\mathrm{Pn}{=}0.6$, $\theta^\star{=}(0.5, 2, 1, 1)$; initialisation $\mathrm{Ko}{=}1$, $\mathrm{Bi}_t{=}0.5$; 20 seeds, 1% noise). The reduced model is frozen on every grid and at every noise level.

forward model	truth / instrument	Ko (true 2.0)	Bi_t (true 1.0)
full ($\mathrm{Pn}{=}0.6$)	matched grid	$2.00{\pm}0.01$ (**0.1%**)	$1.00{\pm}0.00$ (**0.0%**)
reduced ($\mathrm{Pn}{=}0$)	matched grid	$1.00{\pm}0.00$ (frozen)	$0.50{\pm}0.00$ (frozen)
full ($\mathrm{Pn}{=}0.6$)	$4\times$ refined grid	$2.01{\pm}0.01$ (0.6%)	$1.00{\pm}0.00$ (0.1%)
full ($\mathrm{Pn}{=}0.6$)	gain/bias error	$1.75{\pm}0.01$ (12.7%)	$0.98{\pm}0.00$ (2.2%)

Cramér-Rao against Monte Carlo. Table 1 reports 19.2% on Ko where Table 2 recovers it to 0.1%. These agree: the former is the *marginal* bound with $\mathrm{Lu}, \mathrm{Bi}_m$ also unknown, while the estimator holds them fixed. The comparable *conditional* bound is 0.4% against a Monte Carlo spread of 0.37%, which is what validates the Fisher machinery the rest of the paper rests on.

Where the calibration error goes. At the top of the silent band the misfit is absorbed as bias in the parameters the reduced model can see: Lu is pulled from 0.5 to 0.33, the Jacobian's sensitive subspace rotates by $68°$, and the sensitivity ordering inverts – so the calibrated model is wrong in a way no residual reveals.

A withheld instrument. Predicting a channel never fitted separates physical recovery from compensation within the moisture channels. From the moisture-only fit the full model reproduces the surface-temperature trace $T(0, \tau)$ to 0.0% on a matched grid and 2.1% under the instrument misspecification of Table 2; the reduced model, frozen at its initialisation, misses it by 48.9%.

The observation operator deserves equal scrutiny. Writing the mass-flow meter as $\partial_\tau M(0, \tau)$ rather than the conservation-form flux of Eq. 2 leaves the null untouched but inflates apparent

discretisation error by two orders of magnitude. Relatedly, the gain row of Table 1 is non-monotonic because it is a ratio of two bounds improving at different rates: the collinearity collapse near $\mathrm{Pn} \approx 0.8$ helps the moisture-only set far more than the already well-conditioned moisture-plus-$\langle T \rangle$ set.

The joint problem with the coupling known. App. D's audit ends with the dropped coefficient measured, so the practical case is the four-parameter problem at *known* Pn. Recovering $(\mathrm{Lu}, \mathrm{Ko}, \mathrm{Bi}_t, \mathrm{Bi}_m)$ from the two moisture channels under the full model, with Table 2's estimator and 4 random starts from the box per seed (20 seeds), *every* start reaches the same basin in every cell of Table 3: the multimodality of the five-parameter problem below comes from estimating Pn itself, not from the coupled model. Accuracy, though, is strongly graded in Pn and tracks the marginal CR of Table 1. At $\mathrm{Pn}{=}1$ all four recover to a few per cent; at the canonical $\mathrm{Pn}{=}0.6$ benchmark Ko and Bi_t reach only 13% and 20%; at $\mathrm{Pn}{=}0.2$ they are not usefully determined (36%, 48%). Every median estimate is interior to the box, so this is the collinearity of Sec. 4 rather than a boundary artefact. Dissolving the null is therefore not a claim that the moisture-only cell pins down the whole parameter vector: it does so only at $\mathrm{Pn}{=}1$, the strongest coupling we tested, which is the other half of the scoping in Sec. 7. The assumed range runs to $\mathrm{Pn}{=}2$, which we did not test.

Table 3: Joint recovery of all four parameters from the two moisture channels with Pn known and fixed (full model, 20 seeds $\times$ 4 random starts from the box, best-cost start per seed). Median relative error per parameter; "all four" is the fraction of seeds within 10% on every parameter simultaneously. Every start reached the same basin in every row.

Pn	noise	Lu	Ko	Bi_t	Bi_m	all four
0.2	0.5%	2.5%	24.9%	28.1%	4.2%	15%
0.2	1%	4.8%	35.8%	47.6%	5.0%	10%
0.2	2%	8.0%	39.9%	50.0%	8.3%	5%
0.6	0.5%	11.7%	9.5%	13.7%	12.6%	40%
0.6	1%	17.0%	13.2%	19.7%	18.7%	25%
0.6	2%	31.0%	16.9%	45.2%	39.8%	15%
1.0	0.5%	1.4%	0.5%	1.0%	0.9%	100%
1.0	1%	2.8%	1.1%	2.0%	1.8%	100%
1.0	2%	5.5%	2.2%	3.9%	3.6%	90%

The joint problem with the coupling unknown. Recovering all five of $(\mathrm{Lu}, \mathrm{Ko}, \mathrm{Bi}_t, \mathrm{Bi}_m, \mathrm{Pn})$ from random initialisations across the box, the coupling is best determined (Pn to 1.8% median) and Ko reaches 18%, but Bi_t and Bi_m land at 45% and 41% and the multi-start residual costs reveal more than one basin.

C Tested loss and architecture modifications do not restore the severed pathway

If the residual is built from the reduced operator, Ko and Bi_t appear only in the temperature equation and the moisture balance is autonomous. Differentiating that residual with respect to Ko shows the sensitivity $\partial M / \partial \mathrm{Ko}$ obeys the same homogeneous equation with homogeneous initial and boundary data, so its unique solution is zero: the null is a property of the hypothesis space, not of the objective, and a loss can only reweight paths that already exist. A trained network realises this only up to its own residual error, so what is measured is a *leakage floor* rather than an exact zero.

This appendix exists to confirm that analytic claim on a finitely-trained network, not to benchmark mitigations: every arm trains on the reduced operator, so every arm is *expected* to fail, and an arm doing better here would mean its optimisation left less residue, not that it recovered the severed physics. Data from the coupled system, or an embedded discrepancy term, would reintroduce the missing pathway and is the actual fix – that is the point of the mitigation in Sec. 7 – and is out of scope here by construction.

We test it against the remedies a PINN-literate reader would reach for. Five arms train on the same reduced physics with the same budget (12k Adam then 800 L-BFGS steps, width 96, depth 5, 4000 collocation points, three seeds) and differ only in objective or architecture; the last is deliberately the most favourable case, a shared trunk whose moisture head receives the temperature head's output and its spatial gradient as explicit inputs.

Table 4: None of the tested objectives or architectures restores the severed pathway. All arms train on the *same* reduced ($\mathrm{Pn}{=}0$) physics and differ only in objective or architecture; mean $\pm$ s.d. over 3 seeds. The solver's reduced-model value for $\|\partial\langle M\rangle/\partial \log \mathrm{Ko}\|$ is exactly 0; under the true model ($\mathrm{Pn}{=}0.6$) it is 0.372. "Leakage" is the severed-pathway sensitivity as a fraction of the intact $\mathrm{Ko}\to\langle T\rangle$ pathway of the same network, given in the preceding column. †: one of the three baseline seeds does not converge (per-seed leakage 1.8, 7.3, 137.5%), so its mean would not summarise a stable quantity and its denominator is itself unreliable; the median is given instead.

arm	fwd rel-L_2	$\|\partial_{\log \mathrm{Ko}}\langle M\rangle\|$	$\|\partial_{\log \mathrm{Ko}}\langle T\rangle\|$	leakage	vs. true
baseline residual	0.066 ± 0.072	0.233 ± 0.202	1.52	$7.3\%^{\dagger}$	63%
+ gradient-enhanced penalty	0.008 ± 0.004	0.121 ± 0.075	2.16	$5.8 \pm 3.8\%$	32%
+ gradient-norm balancing	0.008 ± 0.003	0.027 ± 0.018	2.33	$1.1 \pm 0.7\%$	7%
+ causal weighting	0.013 ± 0.005	0.115 ± 0.064	2.09	$5.7 \pm 3.4\%$	31%
shared trunk, T and $\partial_\xi T$ to the M head	0.010 ± 0.002	0.123 ± 0.049	2.14	$5.9 \pm 2.5\%$	33%

Reading. The structural claim is the analytical one above, and it is exact: at convergence the reduced residual forces $\partial\langle M\rangle/\partial\mathrm{Ko}$ to zero, so no objective can reintroduce the pathway. The table reports what a *finitely*-trained network does with that, over three seeds.

Three things follow. First, leakage is optimisation residue, not a structural quantity: the plain baseline is unstable across seeds (1.8, 7.3, 137.5%; one run fails to converge, so Table 4 gives its median), while every modification that stabilises training also drives leakage down, gradient-norm balancing furthest ($1.1 \pm 0.7\%$). Second, and the point of the experiment, *none* of them moves the severed sensitivity toward the true coupled value; they move it toward zero, which is the correct answer under the reduced physics. Third, the architecture built to let the pathway reappear buys nothing: the shared trunk, which receives T and $\partial_\xi T$ as explicit inputs, sits at $5.9 \pm 2.5\%$ against $5.8 \pm 3.8\%$ for a gradient-enhanced penalty and $5.7 \pm 3.4\%$ for causal weighting – indistinguishable from modifications with no such design, and beaten by pure loss balancing. The practical consequence is why the diagnostic of Sec. 4 runs on the solver and not the surrogate: even at the best arm the residue is 7% of the true coupled sensitivity, so a surrogate's tangent map can report a coupling *small* but cannot certify it *zero*. The loss column is omitted as it is not comparable across arms, the adaptive arm reweighting the boundary term by construction.

D A three-step audit for a candidate reduction

The mitigation of Sec. 7 is a procedure, and this is it. It is cheap: every step below is a solver call, and the whole audit for our system runs in minutes on a laptop CPU.

1. Enumerate what the reduction dropped, and what it severs. For each term the reduced model omits, write down which directed paths it removes from the observation operator $H(\theta)$ *and* from $H(u)$ – the two differ, and a reduction safe for inference can still be fatal for control, as $\mathrm{Pn}{=}0$ is here. For the Luikov reduction, dropping $\mathrm{Lu}\,\mathrm{Pn}\,\partial_{\xi\xi}T$ severs the only path from $\{\mathrm{Ko}, \mathrm{Bi}_t\}$ to the moisture channels, and the only path from u to $\bar{M}$.

2. Probe the crossover at the bottom of each plausible range. Restore the term in the simulator, run one probe solve at a small coefficient value, and read off $\mathrm{Pn}^\star \approx \mathrm{CR}(\theta_j; \mathrm{Pn}_{\text{probe}}) \cdot \mathrm{Pn}_{\text{probe}}$ (Sec. 4), with its stated assumptions. Comparing $\mathrm{Pn}^\star$ against the *lower* end of the coefficient's plausible range is the conservative test.

3. Act on the comparison. If $\mathrm{Pn}^\star$ falls inside the plausible range, the reduced model's verdict is a phantom and must not be used for identifiability or sensor design: replace the implicit $p(\mathrm{Pn}){=}\delta_0$ with a spike-and-slab prior, measure or estimate the coefficient, and re-run the design analysis on the coupled model. If $\mathrm{Pn}^\star$ sits above the whole range, the reduced verdict stands on the evidence available. If the range straddles $\mathrm{Pn}^\star$, report the verdict as range-dependent rather than resolved – and note that goodness of fit will not settle it, since the silent band of Sec. 5 overlaps exactly this region.

E Phantom uncontrollability: closed-loop detail

The control plant of our repository takes the hot-air temperature u as the manipulated variable and advances moisture with an operator that references neither u nor T – the $\mathrm{Pn}{=}0$ reduction. Its

docstring nonetheless asserts that higher air temperature dries faster, which the implementation makes impossible: the control-side instance of this paper's thesis, authored by us. Its input-to-moisture Jacobian is therefore exactly zero, not merely small (Fig. 2a), while the actuator retains full authority over surface temperature ($\|\partial T(0,\tau)/\partial u\| = 9.0$). Restoring the coupling recovers drying authority at first order, with measured log-log slopes $+0.994$ ($\bar{M}$) and $+0.995$ (drying rate) for $\mathrm{Pn} \leq 0.1$ – the same law the parameter-to-observation Jacobian obeys.

Task, objective and arms. Reach a terminal mean-moisture target while holding the surface at a quality setpoint below a scorching limit. Raising u helps the temperature term and hurts the moisture term, a trade-off that exists only for $\mathrm{Pn} > 0$. The shrinking-horizon cost is

$$J = q_T \sum_k (T_k - T_{\text{set}})^2 \Delta\tau + q_M (\bar{M}_N - \bar{M}_{\text{target}})^2 + r \sum_b (\Delta u_b)^2 + \rho \sum_k \left[T_k - T_{\text{crit}}\right]_+^2 \Delta\tau, \quad (4)$$

with T_{set}=0.55, T_{crit}=0.65, $\bar{M}_{\text{target}}$=0.565, q_T=1, q_M=400, r=0.05, ρ=200; plant step $\Delta\tau$=0.01 over $\tau \in (0, 2]$, a control decision every 10 steps (20 decisions), 3 piecewise-constant blocks spanning the remaining horizon, input bounds $u \in [0, 2]$, and L-BFGS-B (40 iterations, ε=10^{-4}) with 4 starts per solve – multistart, so that a correct prediction model cannot lose to a hedged one merely through a local basin. The three arms differ *only* in the prediction model's coupling; horizon, cost, bounds, solver and tuning are identical, so the independent variable is simulator structure: **reduced** ($\mathrm{Pn}_{\text{model}}$=0), **matched** ($\mathrm{Pn}_{\text{model}}$=$\mathrm{Pn}_{\text{true}}$), and **model-uncertain** (spike-and-slab, mass 0.5 at zero, remainder on $\{0.2, 0.4, 0.6\}$).

Table 5: Closed-loop cost on a coupled truth plant, with 4 multistarts per solve. "Control-sens. T term" is the surface-temperature tracking term, the objective component the reduced model treats as controllable. Rows marked † have the moisture target outside the reachable set at that coupling and are excluded from the regret claim.

Pn (truth)	reduced J	matched J	uncertain J	regret	u red./matched	control-sens. T
0.10†	1.007	0.898	0.911	1.1×	1.18/1.23	0.025
0.20	0.060	0.042	0.087	1.4×	1.16/1.19	0.022
0.35	0.751	0.094	0.093	**8.0**×	1.13/0.87	0.019
0.50	3.103	0.261	0.246	11.9×	1.11/0.63	0.017
0.60	5.267	0.420	0.403	12.5×	1.09/0.52	0.016
0.80†	10.460	0.924	0.884	11.3×	1.07/0.35	0.016

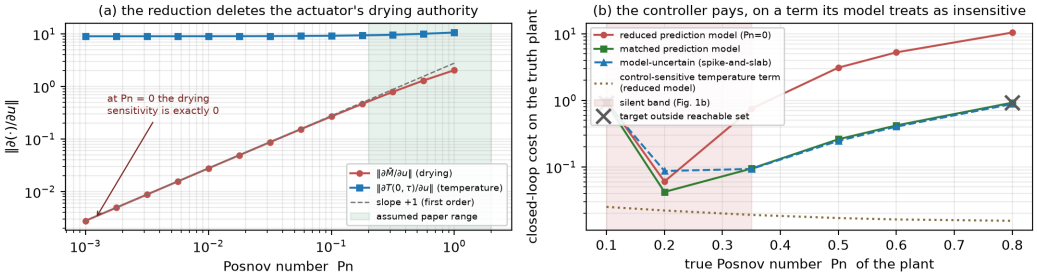


Figure 2: **(a)** Sensitivity of the drying and temperature observables to the manipulated air temperature. Exactly zero for moisture at Pn=0, $O(\mathrm{Pn})$ thereafter; the temperature channel is flat and large throughout. **(b)** Closed-loop cost on the coupled truth plant. The reduced-model controller's regret grows with the true coupling while the control-sensitive temperature term under the reduced model (dotted) improves.

Reading. Three points. First, this is a *decision reversal*: at Pn=0.6 the matched controller runs less than half the reduced controller's mean heat (0.52 vs 1.09), because the reduction hides that cooling is the way to dry. Second, the failure is silent in the same sense as Sec. 5 – the control-sensitive temperature term under the reduced model improves monotonically ($0.025 \to 0.016$) as its true cost rises by an order of magnitude. Third, Pn=0.35 lies inside the silent goodness-of-fit band, so a model that cannot be rejected by fitting the moisture sensors still costs 8× in closed loop, there

commanding a 30% higher mean thermal input than the matched controller. Model-uncertain control tracks the matched arm for $\mathrm{Pn} \geq 0.35$, within $\pm 5\%$ of it – comparable rather than better – while costing substantially more at $\mathrm{Pn}{=}0.2$, where the coupling is too weak to be worth hedging against. Every solve uses 4 multistarts; against a single warm start no closed-loop cost moves by more than 0.05%, so these differences are not local-basin artefacts.

Idealisations. Full-state feedback (no observer), a solver-in-the-loop shrinking-horizon NMPC rather than a learned surrogate, and no mapping from dimensionless time to a physical sampling interval, so no real-time claim is made. These deliberately remove estimator and surrogate quality as confounds. The coupled plant is bit-identical to the original reduced plant at $\mathrm{Pn}{=}0$ over 150 random control steps, and reproduces the benchmark solver to $0.00\mathrm{e}{+00}$ relative L_2 at constant input.